

SYNAPSE

The Teacher's World

Every page. Every feature. One workspace.

An AI workspace and classroom command center for Zambian teachers — planning, assessment, communication and evidence, all aligned to the official curriculum and all carrying the school's name.

The promise: hours returned every week, documents worthy of inspection, and a classroom that teaches the teacher what to do next.

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1. What the Teacher's World Solves

Zambian teachers carry heavy loads: large classes, full syllabus pacing, MoE paperwork, parent expectations — often with limited materials. The Teacher's World attacks the three biggest time sinks directly:

Time sink today	How Synapse answers it
Planning & formatting (2–4 hrs/week)	AI generates complete MoE-format lesson plans and schemes from curriculum dropdowns; voice-to-plan; regenerate any section.
Setting & marking assessments (3+ hrs/week)	AI builds quizzes, tests and ECZ-format exams with marking guides; AI marks submissions; teacher verifies in taps.
Parent & class communication (scattered across WhatsApp)	One official channel: announcements, class groups, parent messages, meetings — logged and branded.

2. The App at a Glance — Eleven Pages

Page	Purpose
1. Home (Today)	Everything awaiting the teacher today, at a glance
2. Plan	Lesson plans, schemes of work, voice-to-plan, curriculum browser
3. Assess	Quiz/test/exam builder, question bank, AI marking, gradebook
4. Class	Classes & codes, learners, mastery entry, attendance-lite
5. Insights	Class analytics, mastery heatmap, adaptive recommendations
6. Connect	Announcements, class groups, parent messaging, meetings
7. Documents	Branded PDFs, report cards, certificates, print center
8. Library	My resources, school-shared plans, upload review queue
9. Grow	Observations, teaching standards, professional development
10. Calendar	Term planner, deadlines, events, reminders
11. Me	Profile, school branding preview, settings, offline

3. Page One — Home (Today)

- Today's lessons — tap any to open its plan; time till each period.
- Action queue — mastery scores waiting to be entered, submissions awaiting verification, review-queue items pending. Nothing gets forgotten; nothing needs a notebook.
- Quick actions — new plan, new quiz, announce, mark attendance.
- Pulse card — class average mastery this week, trending up or down, and the one competence most needing attention tomorrow.
- School announcements and parent messages — priority first, quiet by default.

4. Page Two — Plan (AI Planning)

4.1 Lesson Plans

- Cascading curriculum dropdowns: grade → subject → topic → sub-topic → specific competence — official CDC data, so the curriculum part can never be wrong.
- One tap generates the complete MoE-format document: general and specific competences, lesson goal, rationale, prior knowledge, references (Zambia Education Curriculum Framework 2023), learning environment, materials (realistic and low-resource aware — charts, manila paper, textbooks), expected standard, and the 5-stage progression table (Introduction → Development → Exercise/Assessment → Homework → Conclusion) with Teacher activity / Learner activity / Assessment criteria columns.
- Edit inline; regenerate any single section; version history; duplicate across classes; export branded PDF.

4.2 Schemes of Work

- Fill Week 1 — the engine continues in syllabus order across the term; mark any week Revision or Test; the full term's MoE-format table generates automatically.

4.3 Voice-to-Plan

- Speak the lesson aloud — in English or a local language ("tomorrow I want to teach web interactivity, they struggled last week, practical session") — Synapse structures it into the official format. Talking is faster than typing.

4.4 Smart extras

- Differentiation: the same plan rendered at three support levels (core / supported / advanced) for mixed-ability classes.
- Absent-learner recap: a one-page plain-language summary of each lesson for students who were absent — generated automatically, sent to their Planner.
- Curriculum browser: the whole syllabus as a tap-through map — see what was taught, what's next, and how every lesson connects.

5. Page Three — Assess (Quizzes, Tests, Exams, Marking)

5.1 Quiz & Test Builder

- Pick a competence → AI generates a quiz (multiple choice, structured, calculation, short essay) with answers and mark allocations; adjust count and difficulty with sliders.
- Diagrams where needed: LaTeX-rendered mathematics, real music notation, generated figures for sciences and technical subjects.
- Push to the class in one tap — it appears in every learner's Planner with a due date; parents are notified automatically.

5.2 Exam Generation (ECZ-format)

- Full exams learned from the real ECZ paper bank: authentic section structure, command words, mark distribution and time allowance — with a complete marking guide.
- Per-student variants: the same exam with randomized numbers so answer sharing is useless.
- Questions inside a live exam are solution-locked for the whole class until the due date; every solution view is logged for the integrity audit.

5.3 AI Marking & Gradebook

- Submissions marked against the guide; the teacher sees the AI's marking with the reasoning and overrides in one tap.
- Gradebook auto-built from every quiz, test and mastery entry — no spreadsheet, ever.
- Comment writer: per-learner report comments drafted from performance data; the teacher edits tone and detail, never starts from a blank page.

6. Page Four — Class

- Create class → get a class code; learners join in seconds. Multiple classes, multiple grades — one view.
- Mastery entry in under two minutes: after a lesson, mark each learner mastered / not yet for that competence — one screen, big tap targets.
- Attendance-lite: present / absent in seconds; absence triggers a guardian notification.
- Class list with guardian-link status: instantly see which learners have a connected parent and which don't — share a join code with the unlinked.
- Learner snapshots: any learner's pathway — mastery history, strengths, needs work, app activity — one tap from the class list.

7. Page Five — Insights (Analytics & Adaptation)

- Mastery heatmap: the whole syllabus for the class — green (mastered), amber (partial), red (needs work) — at competence-code level.
- "Who needs what" list: per learner, the exact competences to revisit — with one-tap revision packs generated for each.
- Adaptive engine: below-threshold class mastery → the next lesson plan auto-opens with remediation; strong mastery → enrichment. The class teaches the plan; the plan adapts to the class.
- Trends: week-over-week movement, comparison to the school average, early warning before marks slip.
- Integrity audit: which students viewed solutions before a test, and when — evidence, not suspicion.

8. Page Six — Connect (Communication)

- Announcements: to one class, several, or the whole school; parents receive them in their own feed; read receipts show who has seen it.
- Class groups: moderated chat with the learners of a class — questions after school, homework clarifications — with the teacher in control and everything logged (no more WhatsApp groups with no oversight).
- Study-group visibility: see the school-scoped study groups your learners joined and drop in to answer — teaching extends beyond the classroom walls.
- Parent messaging: direct, logged conversations; canned updates ("your child mastered 4.1.4 this week") sent in two taps.
- Meetings: publish available slots; parents book; reminders to both sides. No phone-call tag.

9. Page Seven — Documents (Branded & Print-Ready)

- The school's identity on every page: logo, colors, school name and EMIS header on lesson plans, schemes, tests, report cards and certificates — print-ready PDFs that stand up in inspections.
- Report cards: the term's reports auto-composed from mastery data, per learner, with AI-drafted comments the teacher edits in one tap.
- Certificates: achievement certificates for competitions, perfect scores, improvement — generated and printed in seconds.
- Print center: batch-export anything, formatted for the school's printer.

10. Page Eight — Library (Resources & Review Queue)

- My resources: every plan, scheme, quiz and document the teacher has made — searchable, reusable, versioned.
- School-shared library: colleagues' plans rated and organized by topic; the school's collective knowledge stays even when a teacher transfers.
- Upload anything: a past paper, textbook chapter, or notes → auto-digitized and tagged to competences. Low-confidence items land in the review queue — one tap confirms or fixes. Teachers curate the platform in seconds and the whole system gets smarter.

11. Page Nine — Grow (Professional Development)

- Lesson observations: heads of department record observations against the teaching standards, linked to the actual lesson plan taught.
- Standards tracker: the teacher's growth across the term, with suggested focus areas and relevant resources.
- Micro-PD: short, targeted tips generated from the teacher's own analytics ("your assessments skew recall — here are two question styles for analysis").

12. Page Ten — Calendar

- Term planner: schemes, tests, assignments and events on one timeline; drag to reschedule; learners and parents get deadline reminders automatically.
- Exam-countdown views per class; conflict warnings (two tests same day).

13. Page Eleven — Me

- Profile, subjects and classes, school branding preview, language preference, notification controls, data & privacy settings, offline downloads.

14. How the Pages Synapse (Work Together)

From → To	What flows
Class scores → Plan	Mastery results reshape the next lesson automatically
Plan → Documents	Every plan exports school-branded, print-ready
Assess → Connect	Marked results notify parents with context in two taps
Library → Assess	Reviewed uploads enrich the question and exam bank for everyone
Insights → Grow	Analytics suggest the teacher's own next development step
Calendar → Home	Deadlines surface as today's action queue
Class → Connect	Guardian-link gaps surface so no parent stays unconnected

15. The Workload Ledger (What a Week Looks Like)

Weekly task	Before Synapse	With Synapse
Lesson plans (5)	~3 hours writing & formatting	~15 minutes reviewing AI drafts
Scheme of work updates	~1 hour retyping tables	Automatic from Week 1 onward
Quiz/test creation + marking	~3 hours	~30 minutes verifying AI marking
Report comments	~2 hours from a blank page	~20 minutes editing drafts
Parent updates	Ad-hoc, untracked	Two-tap canned updates, logged
Record keeping	Spreadsheets, paper	Automatic gradebook & mastery
Total	≈ 10+ hours of paperwork	≈ 1.5 hours of oversight

The recovered time goes where it belongs: teaching, feedback, and the learners who need it most.

16. Built On

- Curriculum engine (CDC syllabi, official codes), knowledge bank (auto-digitized ECZ papers), adaptive mastery loop, branded documents, and the same security model as the whole Synapse platform (row-level privacy, minors protected).
- Technical base: Flutter + Supabase + LLM functions — the Teacher's World is one role in the same app learners and parents use.

17. First Build (MVP Slice)

- Pages 1-7 with: plans, schemes, quiz builder, class codes, mastery entry, adaptive adjustment, branded PDFs, parent messaging. Everything else follows in phases — the schema already supports it all.
- Open dependency, unchanged: one CDC syllabus PDF to digitize and ECZ papers to ingest. Real content in, real hours saved out.